

2. Iris Dataset

Load the datasets:

```
load("data/iris.train.rdata")  
load("_data/iris.test.rdata")
```

Create a new folder named “_functions” and load the function:

```
source("_functions/CV.class.R")
```

Load the packages,

```
library(MASS)
```

- a) Consider the famous iris dataset `iris.train.rdata` introduced in the lectures. Reproduce the pairs plot for the four sepal and petal variables as given in the lectures. Which variable appears to be discriminating the species best? And which is worst?
- b) Explain the difference between “discrimination” and “classification”.
- c) Explain what is meant by the assumption “We assume a priori that versicolor and virginica are equally likely”.
- d) Fit an LDA model to the iris data using `Sepal.Length` as the predictor. Assume equal prior probabilities for both species. Print you fitted model. What are the sample means of each species for this predictor variable?
- e) ~~Source in the `CV.class.R` file (open the file in the script window and press the “Source” button to the upper right). Look at the `CV.class.examples.R` file for reference.~~ Perform a Leave-One-Out Cross-Validation of the model you fitted in the previous exercise. Report the confusion matrix, the accuracy and the cross-validated error rate.

- f) Use the scheme from exercises d. and e. to identify a good classifier for iris species. You may use either `lda` or `qda` and you may use one or several predictors. Report the cross-validated error for you “best choice”.
- g) What is the model assumption difference between an LDA and a QDA model?
- h) Use the model of your choice to predict the samples in `iris.test.rdata`. Use the `confusion()`-function in the `mixlm`-library to evaluate the performance of your classifier.

- Optional
- i) Use a logistic model of your choice (perhaps the same predictors as you used in your best choice classifier of part a) to estimate the posterior probabilities of “virginica” for the samples of the `iris.test.Rdata` set. Allocate the samples to the most probable species class and use the `confusion()` function to evaluate the classification performance. (Hint: The posterior probabilities, say you name them `postprob`, should be classified into a factor variable by

```
predicted <- factor(ifelse(postprob > 0.5, "virginica", "versicolor"))
```